

CNN + CBAM + Grad-CAM

Explainable AI for Pancreatic Tumor Segmentation

Dataset Medical Decathlon — Pancreas (Task07)	Model CNN + CBAM	Explainability Grad-CAM + Explainable AI
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1. Project Introduction

The aim is to build a deep learning system for pancreatic tumor segmentation from CT images. The system uses CNN for feature learning, CBAM to focus on important features, and Grad-CAM as the explainability method. The final system is therefore **CNN + CBAM + Grad-CAM + Explainable AI**.

2. Dataset & Target

Dataset	Medical Decathlon — Pancreas (Task07)
Input	3D CT scans (.nii.gz)
Original labels	0 = Background · 1 = Pancreas · 2 = Tumor
Target	Binary tumor segmentation mask

3. Preprocessing Pipeline — Required Steps

Use all six preprocessing steps below before training the CNN + CBAM model.

01 ROI Patch Extraction

Extract fixed-size 3D patches centered on the pancreas bounding box. This focuses the model on the relevant region and reduces memory use.

02 CLAHE Enhancement

Use Contrast Limited Adaptive Histogram Equalization to improve local contrast and make pancreatic boundaries easier to see.

03 Targeted Zoom Augmentation

Apply scale-aware zoom around the pancreas region using 0.7–1.4× magnification to simulate anatomical size variation.

04 Elastic Deformation

Apply smooth spatial warping with random displacement fields to mimic natural organ deformation and improve generalization.

05 Multiscale Training

Train using 32³, 64³ and 128³ patches so the model can learn both global context and fine local features.

06 Attention-Guided Augmentation

Use saliency maps from a teacher model to identify important regions and concentrate augmentation in those regions.

4. Core Model

CNN	CBAM	Segmentation	Explainability
Learns image features	Channel + spatial attention	Produces tumor mask	Grad-CAM shows important regions

Final system: CNN learns features → CBAM focuses on useful features → segmentation head predicts the tumor → Grad-CAM explains which regions influenced the prediction.

5. Optional Attention / Context Block

If an extra attention/context block is required, choose **one**: Multi-Scale Cross-Attention OR Swin Transformer Block.

Option	Purpose
Multi-Scale Cross-Attention	Combines information from different feature scales.
Swin Transformer Block	Uses window-based transformer attention to capture feature relationships.

6. Fixed Training Setup

Setting	Value	Setting	Value
Cross-validation	5-fold	Epochs	10 per fold
Batch size	8	Optimizer	Adam
Loss	BCE + Dice Loss	Training unit	3D patches / multiscale

7. Bayesian Optimization Search Space

Hyperparameter	Search range	Hyperparameter	Search range
Learning rate	1e-5 – 3e-4	Attention heads	4 – 12
Embedding dimension	128 – 512	CBAM reduction ratio	8 – 32
Attention dropout	0.1 – 0.4	Batch size	2 – 8
Boundary loss weight	0.1 – 0.4	Requested final batch	8

8. 5-Fold Cross-Validation

Step	What to do
1	Split the data into 5 patient/volume-level folds.
2	Use 4 folds for training and 1 fold for validation.
3	Repeat until every fold has been used once for validation.
4	Train each fold for 10 epochs with batch size 8.
5	Save the metrics and checkpoint for every fold.
6	Report mean $\pm$ standard deviation across the 5 folds.

Important: Keep all slices/patches from the same patient or 3D volume in the same fold. This prevents data leakage.

9. Evaluation Metrics

Metric	Simple meaning
Dice	Overlap between predicted tumor and true tumor.
IoU	Intersection over Union of prediction and ground truth.
Precision	How much of the predicted tumor is correct.
Recall	How much of the actual tumor was found.
F1 Score	Combines precision and recall.
Accuracy	Overall pixel-level correctness.

10. Grad-CAM + Explainable AI

Grad-CAM is the explainability method used with the model. Generate a Grad-CAM heatmap for predictions to show which regions of the CT input contributed most to the model's decision. Save the original image, prediction mask, and Grad-CAM visualization together for qualitative analysis.

11. Final Handoff Checklist

- Medical Decathlon Pancreas Task07
- ROI patch extraction
- CLAHE enhancement
- Targeted zoom augmentation (0.7–1.4 $\times$)
- Elastic deformation
- Multiscale training (32³, 64³, 128³)
- Attention-guided augmentation
- CNN + CBAM segmentation model
- Optional: Multi-Scale Cross-Attention OR Swin Transformer
- 5-fold patient/volume-level cross-validation
- 10 epochs per fold
- Batch size = 8
- BCE + Dice loss and Adam optimizer
- Dice, IoU, precision, recall, F1 and accuracy
- Grad-CAM + Explainable AI outputs

FINAL SYSTEM: CNN + CBAM + Grad-CAM + Explainable AI